QUARK–HAWKING BRIDGE

The Quark–Hawking Bridge

Volume I. Theoretical Core: model class, theorems T1–T15, the asymmetry lock

Version 1.3.0

12 campaigns · 15 theorems · IVP-12 12/12

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Abstract

This volume presents the theoretical core of the Quark–Hawking Bridge (QHB) package, version 1.3.0. The package links two sectors of one confinement story: a Yang–Mills-style mass gap responsible for the impossibility of free color charges, and a thermal anchor given by classical point-vortex dynamics. Every statement belongs to an explicitly delimited reduced model class and is pinned by eight normalization conditions N1–N8; no number in the package claims lattice-QCD status.

The distinctive feature of the exposition is radical honesty. Each claim carries one of five statuses: exact (symbolic proof), measured-law (measurement inside the model class), honest-negative (the negative itself is the result), hypothesis (stated but never asserted as fact) and comparison (external literature value). Every claim is accompanied by a one-line falsifier — the cheapest observation that would refute it.

The centerpiece of v1.3.0 is the asymmetry lock: the proof that under complete matter–antimatter asymmetry the mass gap is permanent and free quarks are impossible as separate asymptotic states. The closure threshold eta_crit = 0.0908885 exceeds the cosmological baryon asymmetry eta_B = 6.1e-10 by a factor of 1.49e8; gap filling is self-limiting, and the combinatorial probability of a macroscopic quark accumulation is below 10^(−1.2e31).

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Примечание: оглавление построено на кодах полей. После редактирования щёлкните по оглавлению правой кнопкой и выберите «Обновить поле», чтобы обновить номера страниц.

1. Introduction and the honesty rules

The QHB package grew out of a simple question: what would a confinement research program look like if it demanded falsifiability of itself as strictly as it demanded results? The answer is the falsifier-first architecture: no statement enters the registry without specifying the observation that would refute it. This order turns even failures — the unresolved detector tail, the substantial escape in the free three-body system — into first-class results with the honest-negative status.

The second rule is reproducibility as a first-class requirement. All campaigns are deterministic: fixed seeds, no wall-clock access, byte-identical rebuilds of the data tables. The twelve-pass IVP-12 audit re-derives every headline number through an independent method — symbolic, 100-digit, quadrature, or algorithmic. Bilingual RU/EN mirrors are checked by the twin control: fourteen pairs of numbers and headers must match across both documentation versions.

The third rule separates levels of knowledge. The exact status means a SymPy proof or a 100-digit mpmath verification; measured-law means a measurement inside the pinned model class with no right to extrapolate beyond the window; honest-negative records the boundary of applicability; hypothesis keeps tempting but unproven ideas away from the results; comparison is reserved for external values that may support consistency but never prove the model. Volume I presents the theory; Volume II presents the campaigns and the audit.

2. The model class and normalization conditions N1–N8

The QHB model class is a dimensionally reduced construction with two sectors. The gap sector is described by the gap coordinate Δ and the string tension σ; the anchor sector by the Kirchhoff equations for three equal-circulation point vortices. The sectors are coupled thermally: the bridge temperature T_c carries the parent project's quantum spin corrections $\pi^2/8$ and $\pi^5/704$. To give every number in the package a single basis, eight normalization conditions are introduced.

Table 2.1 — Normalization conditions

No	Content	Value
N1	Vortex units: Gamma = a = 1	omega = 3/(2pi) = 0.477464829...
N2	Gap units: C9 valley floor	beta* = 3.3583, Delta_F,min = 0.181777

No	Content	Value
N3	Naive-window falsifier	beta_c = 2.9910; measured 6.3x threshold
N4	Detector cutoff	E_max = 2.5
N5	Confinement box	L_box = 12
N6	Cosmological comparison	eta_B = 6.1e-10 (Planck 2018)
N7	Anchored static barrier	V* = 0.9359375
N8	Campaign seeds	20250918 / 777001 / 515003 / 902107

Conditions N1–N8 are conventions, not measurements: they fix the units and reference points against which all subsequent quantities are defined. This is more transparent than hidden constants in code — anyone reproducing the package knows exactly where the conventions end and the computation begins. The detector threshold Delta_thresh = 0.0363485, for instance, is defined through N2 and N3: the gap at the falsified naive point divided by the measured ratio 6.3. Every convention is documented in qhb/constants.py and in docs/theory.

3. The gap sector: finite volume and the phase map

The finite-volume gap is defined as $\Delta_F(\beta, L) = \Delta_F(\beta)\cdot(1 - 0.35/L + 0.03/L^2)$, where the profile is a quadratic valley with the floor at $\beta^* = 3.3583$ and depth $\Delta_{F,\min} = 0.181777$. Theorem T1 states the positivity of the gap over the whole explored region: neither the C1 grid (9 volumes × 9 couplings) nor the C9 phase map (9 volumes × 41 couplings) contains a single CLOSED cell. The C1 grid minimum — 0.174490 at (beta, L) = (3.45625, 6) — stays five times above the detector threshold.

The key honesty caveat concerns the continuum. Since the data is finite-volume ($L \le 32$), the package does not claim the continuum gap Δ_{cont}: the idea is stored in the registry with the single hypothesis status (HYPO-CONT), and its falsifier is honest — an L-extrapolation beyond L = 32. The valley is interior, with the gap growing on both sides in beta, which makes the phase map robust to window shifts.

Naive narrow-window extrapolation serves as a built-in lesson. The exploratory-build extrapolator predicted closure at beta_c = 2.9910; the direct probe measured the gap there at 0.228995 — 6.3 times the threshold. A reproducible second extrapolator (linear least squares on the right flank [3.6, 4.0]) predicts a root at 2.9661 with the ratio 6.48 — also falsified. The lesson is recorded in docs/phase_map: extrapolating across a curved valley produces phantom closure points.

4. String and thermal sectors

The Wilson law W = exp(−σRT) is exact in the model class, and the string tension is measured as a log-slope of loops with a recovery error below 1e-16 (T2). The measured law σ(β) = 0.25·(Δ_F(β)/Δ_F,min)^1.5 ties the string to the gap: at the valley floor σ* = 0.25. Theorem T4 fixes the bridge temperature $T_c = \sqrt{\sigma}/(2\pi)\cdot(1 + (\pi^2/8)\beta^{-2} + (\pi^5/704)\beta^{-4})$: at beta* the total correction is +11.28%, with the first term dominant and the second an order of magnitude and a half smaller.

The coefficients $\pi^2/8$ and $\pi^5/704$ are inherited from the parent project and verified to one hundred digits with mpmath (T3, exact). Symbolic verification is not decoration but part of the reproducibility contract: anyone can repeat it with a single make audit call. The tunneling window of the linear barrier is closed (T9): the action S(0) = 13.8564 gives the transmission T(0) = 9.21e-13 < 1e-12; the window opens only for E < 0.0059. The reduced coordinate carries the effective mass 1/2 — a convention documented next to the formula.

The detector response demonstrates the power of honest negatives. Fitting the pure spectral core on [0, 2.5] leaves sign-constant residuals in the tail zone: all fifty residuals are positive, the runs test is degenerate, and randomness is rejected. The package does not rescue the fit by introducing a tail shape — instead it claims only an amplitude upper bound (0.00035 at 95%). The tail shape beyond E_max = 2.5 remains unresolved; this is T5 with the honest-negative status.

5. Three-body systems: freedom and binding

The free three-body system sets the baseline: without the bridge coupling, escape is substantial. The kinetic energy of six Gaussian components is distributed as chi2_6/8, and the capture threshold is chosen so that the analytic gamma tail equals 0.2433 exactly; the Monte-Carlo with the pinned seed yields precisely 2433 escapes out of 10000 (T6). The analytic-simulation agreement — a z-score within three binomial sigmas — is re-checked by audit pass IVP-05 on every run.

In the bound system the Y-string vertex coincides with the Fermat point: SymPy proves that in the symmetric configuration the angles between the arms are exactly 120 degrees (T7, exact). The spectrum over nine triangle shapes is minimized by the equilateral one — the energy E_Y = sigma·L_Fermat is monotone in the deviation from symmetry (T8). The connection to experimental phenomenology stays at the comparison status: the package models a class, not quarkonium spectroscopy.

The bootstrap closes the statistical circuit (T10). Nonparametric regeneration of the valley floor (2000 resamples, seed 515003) gives a bias of 0.0046 — 0.38 bootstrap errors; the jackknife gives 0.0016. Both are two orders of magnitude below the gap itself,

confirming the robustness of the reference numbers to the estimation procedure. The falsifier is simple: a bias above half the bootstrap error or 5% of the gap would refute T10.

6. The vortex anchor

The Kirchhoff equations for three equal-circulation vortices are implemented from scratch — no external vortex-dynamics libraries. The equilateral triangle is a relative equilibrium rotating at omega = 3/(2pi); an RK4 run with step 1e-4 reproduces the frequency with the deviation 2.3e-15. The corotating-frame linearization spectrum is {0, 0, ±iomega, ±iomega}: the characteristic polynomial $\lambda^2(4\pi^2\lambda^2+9)^2/(16\pi^4)$ is derived by SymPy with zero deviation (T11, exact). The double degeneracy of the oscillatory pair is an exact property, not a numerical accident.

Boundedness follows from the Aref invariants. For $H < 0$ escape is impossible and the maximum pairwise distance is bounded above by $2\cdot\sqrt{I}$ — an inequality provable elementarily. The pinned run (a0 = 1.7, displacement 0.8627, T = 30) measures r_max = 2.7654 against the bound 4.5075 with the invariant drift below 1e-14; the virtual wall L_box = 12 stays far beyond the achieved distance (T12). There is no wall — the boundedness is topological, enforced by invariant conservation.

The anchored slow coordinate s carries the potential $V(s) = V^*\cdot s^2\cdot e^{(2-2s)}$ whose maximum V* = 0.9359375 is attained exactly at s = 1: the derivative $dV/ds = 2V^*e^{(2-2s)}\cdot s(1-s)$ vanishes symbolically in SymPy. All three hundred trajectories with energies in [0.157, 0.865] stay below the barrier — no escapes (T13, 0/300). The falsifier is a single escape with $E < V^*$; its absence is re-checked by every C11 campaign run.

7. The asymmetry lock and the no-free-quarks theorem

The central result of v1.3.0 answers the question of the gap's permanence. The matter sector is modeled as a perturbation eta·U1 with unit sup-norm; the closure-barrier loss is at most 2eta for any profile, because both values of U1 — at zero and at the valley floor — lie in [−1, 1]. The worst case is attained by the step profile and equals exactly 2. Hence theorem T14: the barrier stays positive for all $\eta < \eta_{crit} = \Delta F_{,min}/2 = 0.0908885$. The cosmological baryon asymmetry $\eta_B = 6.1\text{e-}10$ sits 1.49e8 times below the threshold — nine orders of magnitude of margin.

Theorem T15 translates the lock into observable consequences. A free quark has no admissible production channel: the tunneling window is closed (T9), the anchored barrier is impassable (T13), and gap closure is excluded for admissible asymmetries (T14). The observational route is closed as well: were free quarks to exist, they would scatter in the quantum field, screen the linear potential and fill the gaps, forming isolated macroscopic

accumulations. The toy filling dynamics exhibits negative feedback — filling raises the local closure cost — and yields zero saturated wells in a million steps at eta_B, while the positive control at eta = 0.5 cascades.

The combinatorial bound finishes the accumulation scenario: even granting the per-attempt probability 9.2e-13, an object of N ~ 1e30 quarks has probability below 10^(−1.2e31); even thirty simultaneously tunneled quarks give 10^(−361). Experimental searches for free quarks and nuggets bound their fraction at the level of 1e-27 per nucleon — consistent with the zero prediction, but by the package rules this argument carries the comparison status only: it supports consistency, never proves the model. The volume's conclusion: the observed world of bound hadrons is the forced outcome of complete asymmetry, not a coincidence.

8. The theorem registry

The fifteen theorems form three tiers: the symbolic framework (exact), the measured laws of the model class (measured-law), and the honest negatives. The summary below lists the status of each; full statements with uncertainty budgets live in the claims registry data/claims.json and in RESULTS.md.

Table 8.1 — The theorem registry T1–T15

ID	Statement (brief)	Status
T1	Finite-volume gap positive over the whole region	exact
T2	Wilson law exact; sigma(beta) recovered with err < 1e-16	measured-law
T3	Spin coefficients pi^2/8, pi^5/704 — 100 digits	exact
T4	Bridge temperature: correction +11.28% at beta*	measured-law
T5	Detector tail beyond E_max unresolved	honest-negative
T6	Free three-body escape P_esc = 0.2433 ± 0.0043	honest-negative
T7	Y-string vertex is the Fermat point, angles 120°	exact
T8	Y-string spectrum minimized by the equilateral shape	measured-law
T9	Tunneling window closed: T(0) = 9.21e-13	measured-law

ID	Statement (brief)	Status
T10	Bootstrap bias 0.38 boot_se; jackknife agrees	measured-law
T11	Vortex spectrum {0, 0, ±iω, ±iω}, poly $\lambda^2(4\pi^2\lambda^2+9)^2/(16\pi^4)$	exact
T12	Aref boundedness 2√I; r_max = 2.7654; rotation 2.3e-15	measured-law
T13	Static barrier V* = 0.9359375; 0/300 escapes	measured-law
T14	Asymmetry lock: eta_crit = 0.0908885 = Delta_F,min/2	exact
T15	No free quarks: 0/10^6; accumulations < 10^(−1.2e31)	honest-negative

Each row of the registry carries a falsifier and a five-field uncertainty budget: statistical, systematic, model, numerical and external components are accounted separately.